Homework 3

October 15, 2026

Problem 1

Let $C_{n-1}(t)$ be the space of continuous piecewise linear functions on $[0, n-1]$ whose slope is constant on each interval $[i-1, i]$, $i = 1, \ldots, n-1$. Define $\mathscr{A} : C_{n-1}(t) \to C_{n-1}(t)$ by

$$\mathscr{A}[f(t)] = f(n-1-t).$$

(1) Prove that $\mathscr{A}$ is a linear transformation.

(2) Let $\epsilon_j(t) \in C_{n-1}(t)$, $j = 1, \ldots, n$, be the nodal basis satisfying $\epsilon_j(i-1) = \delta_{ij}$ for $i, j = 1, \ldots, n$. Determine the matrix representation of $\mathscr{A}$ with respect to this basis.

Problem 2

Let I_1 and I_2 be the polynomial ideals generated by

$$a_1(\lambda) = (\lambda+1)(\lambda+2)^2, \qquad a_2(\lambda) = (\lambda+2)(\lambda+3)(\lambda+5),$$

respectively. Compute $I_1 + I_2$ and $I_1 \cap I_2$.

Problem 3

Let I_1 and I_2 be the polynomial ideals generated by

$$\begin{aligned} a_1(\lambda) &= \lambda^4 + \lambda^3 - \lambda - 1, \\ a_2(\lambda) &= \lambda^4 + 2\lambda^3 + 3\lambda^2 + 2\lambda + 1, \end{aligned}$$

respectively. Compute $I_1 + I_2$ and $I_1 \cap I_2$.